

Smart EV Charging Network Planning System for Kopargaon Taluka

Team: Team Nirdhar

Domain: Transportation

Hackathon: Smart Kopargaon Hackathon 2026

College: Shri Vile Parle Kelavani Mandal's College of Engineering, Shirpur

Table of Contents

1. Executive Summary
2. Team Details
3. Problem Statement
4. Introduction
5. Ecosystem & Connectivity Maps
6. Objectives & Scope
7. Existing System vs. EVGrid AI
8. Proposed Solution & Uniqueness
9. Innovation Highlights
10. EVGrid AI – Planning Workflow
11. Functional Requirements & UI Layout
12. Non-Functional Requirements
13. System Architecture
14. Database Design
15. Technology Stack Architecture
16. AI Evaluation Parameters & Score Categories
17. Proposed Locations & Field Photographs
18. Expected Outputs & Budget Breakdown
19. User Flow Diagram (App Flow)
20. Implementation Roadmap
21. Risk Analysis & SDG Alignment

1. EXECUTIVE SUMMARY

The increasing adoption of Electric Vehicles (EVs) has created a growing demand for well-planned charging infrastructure. Kopargaon Taluka, a key gateway to Shirdi with high tourist and commuter traffic, currently lacks a scientific approach to planning EV charging stations. Existing methods rely on manual surveys, resulting in inefficient location selection and resource utilization.

EVGrid AI is an AI- and GIS-powered decision support system that identifies optimal charging station locations by analyzing traffic density, population, tourism, road connectivity, electricity grid capacity, parking availability, and future EV demand. The platform provides intelligent recommendations to help government authorities plan charging infrastructure efficiently and cost-effectively.

The solution is expected to improve charger accessibility, optimize infrastructure investment, reduce carbon emissions, and support the transition to sustainable transportation. It also enables long-term planning through demand forecasting and grid feasibility analysis.

Key Differentiator: EVGrid AI integrates AI-based location scoring, GIS mapping, demand forecasting, budget optimization, and government-focused decision support within a single unified platform. Designed initially for Kopargaon Taluka, the system can be scaled to districts, states, and smart cities across India.

2. TEAM DETAILS

Team Nirdhar combines technical research, documentation, presentation, GitHub management, and demonstration to develop EVGrid AI for sustainable development of Kopargaon Taluka.

Sr.	Name	Role	Responsibility
1	Siddhi Kothawade	Team Leader	Report, Planning & Coordination
2	Siddhi Chinchole	Presentation Lead	PPT Design & Research
3	Pranav Chaudhari	GitHub Lead	Repository & Documentation
4	Mayur Patil	Demo Lead	Video Recording & Editing

3. PROBLEM STATEMENT

India is witnessing rapid adoption of electric vehicles due to government incentives, lower operating costs, and environmental awareness. Kopargaon Taluka, located near the pilgrimage town of Shirdi, experiences significant tourist and highway traffic. Despite this growth, EV charging infrastructure is limited and often planned without scientific analysis. Current approaches rely on manual surveys and isolated decisions rather than demand forecasting, GIS analysis, and grid feasibility.

Official Problem Statement:

"The rapid adoption of electric vehicles requires strategically planned charging infrastructure. However, identifying suitable locations based on demand, accessibility, and grid availability remains challenging."

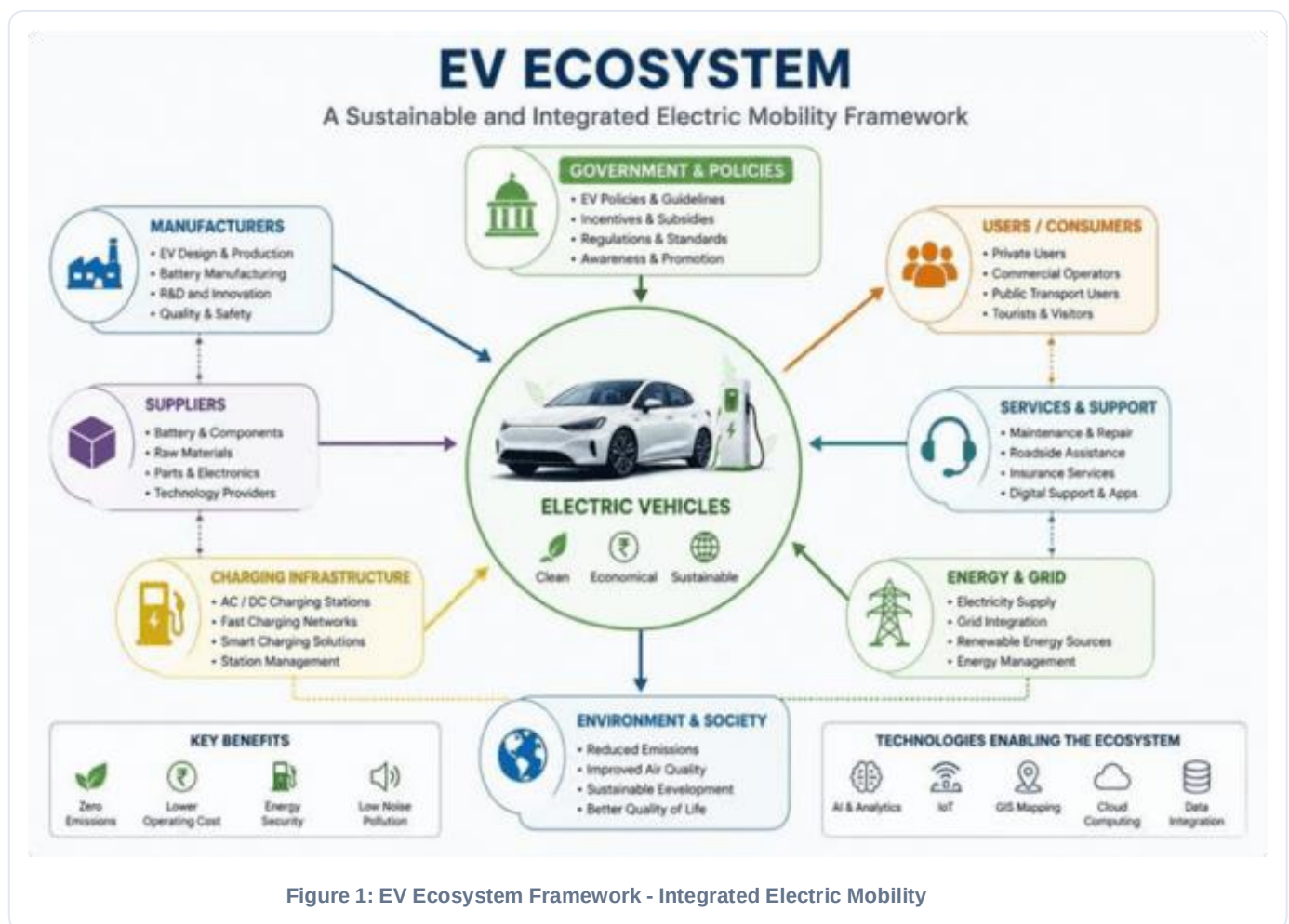
4. INTRODUCTION

Electric Vehicles (EVs) are powered wholly or partly by electricity and play a major role in reducing fossil fuel consumption and greenhouse gas emissions. India's transition toward electric mobility is accelerating through national and state policies. Kopargaon Taluka is well positioned as a pilot region because of its strategic location, connectivity to Shirdi, growing urbanization, agriculture, commerce, and increasing visitor movement.

EVGrid AI integrates Artificial Intelligence, Geographic Information Systems (GIS), and spatial analytics to identify optimal charging locations, estimate future demand, and support budget planning. Major factors considered include population density, tourist movement, road connectivity, electricity substations, parking availability, hospitals, schools, markets, and future urban expansion.

5. ECOSYSTEM & CONNECTIVITY MAPS

Below are the spatial connectivity routes for Kopargaon Taluka to Shirdi and the comprehensive EV Mobility Ecosystem Framework.



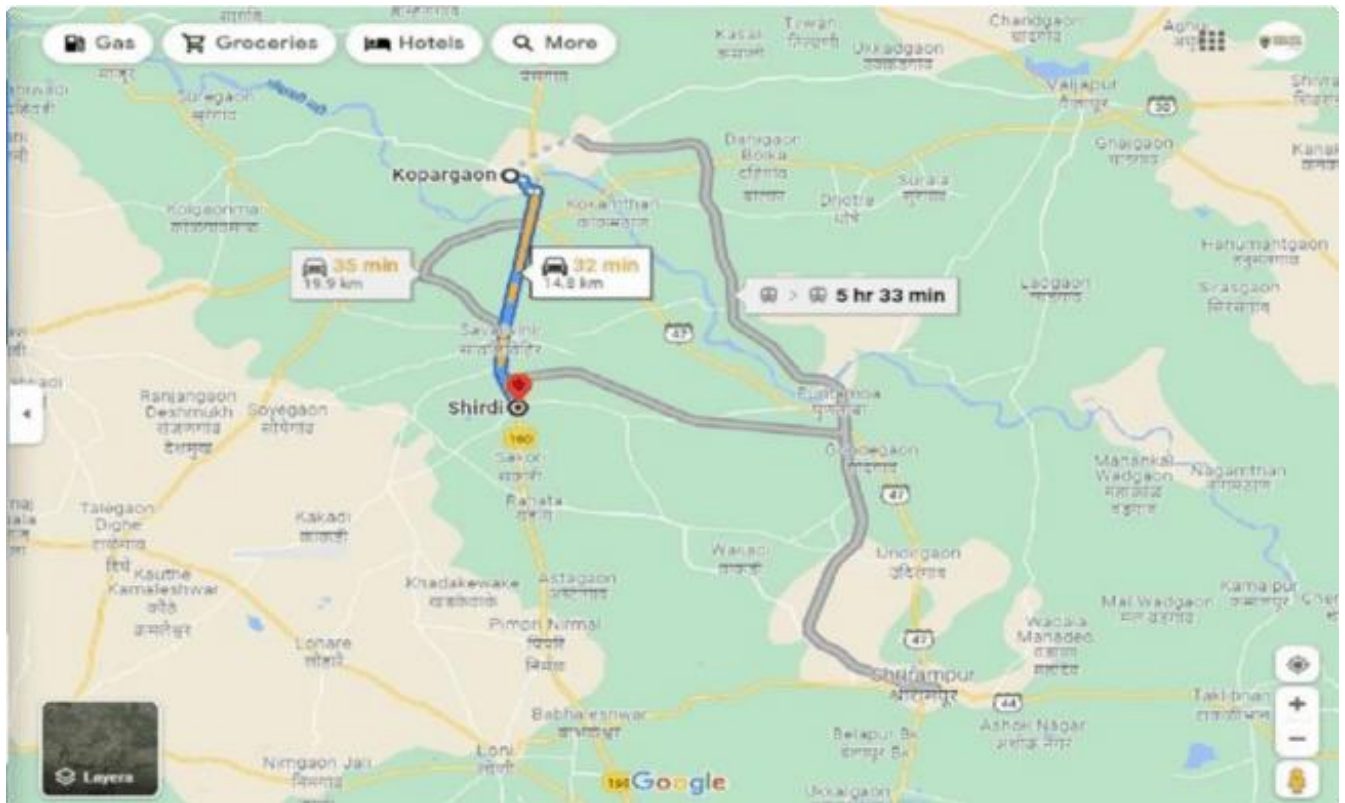


Figure 2: Spatial Route & Connectivity Map (Kopergaon Taluka to Shirdi)

6. OBJECTIVES & SCOPE

Objectives

- Reduce charging accessibility issues and range anxiety
- Optimize charging station placement using AI and GIS
- Support sustainable transportation and reduce carbon emissions
- Assist government authorities in efficient infrastructure planning
- Improve utilization of public resources and grid infrastructure
- Promote EV adoption in Kopergaon Taluka

Scope

The project will initially focus on Kopergaon Taluka as a pilot implementation. The platform is designed modularly so it can be expanded seamlessly to the entire Ahmednagar District, Maharashtra State, and Smart Cities across India.

7. EXISTING SYSTEM VS. EVGRID AI

Current EV charging infrastructure planning mainly depends on manual surveys, traffic observation, and departmental approvals. These methods require significant time and often fail to account for future EV growth, tourism demand, and spatial grid analysis.

Existing Approach	Key Limitation
Manual Surveys	Time-consuming, labor-intensive, and prone to human error
No GIS Integration	Poor spatial location analysis and geographic visualization

Existing Approach	Key Limitation
No AI Analytics	Lacks smart recommendations and multi-parameter site ranking
No Demand Forecasting	Cannot predict future EV adoption trends or grid bottlenecks
Siloed Planning	Uneven charging coverage and inefficient deployment of public funds

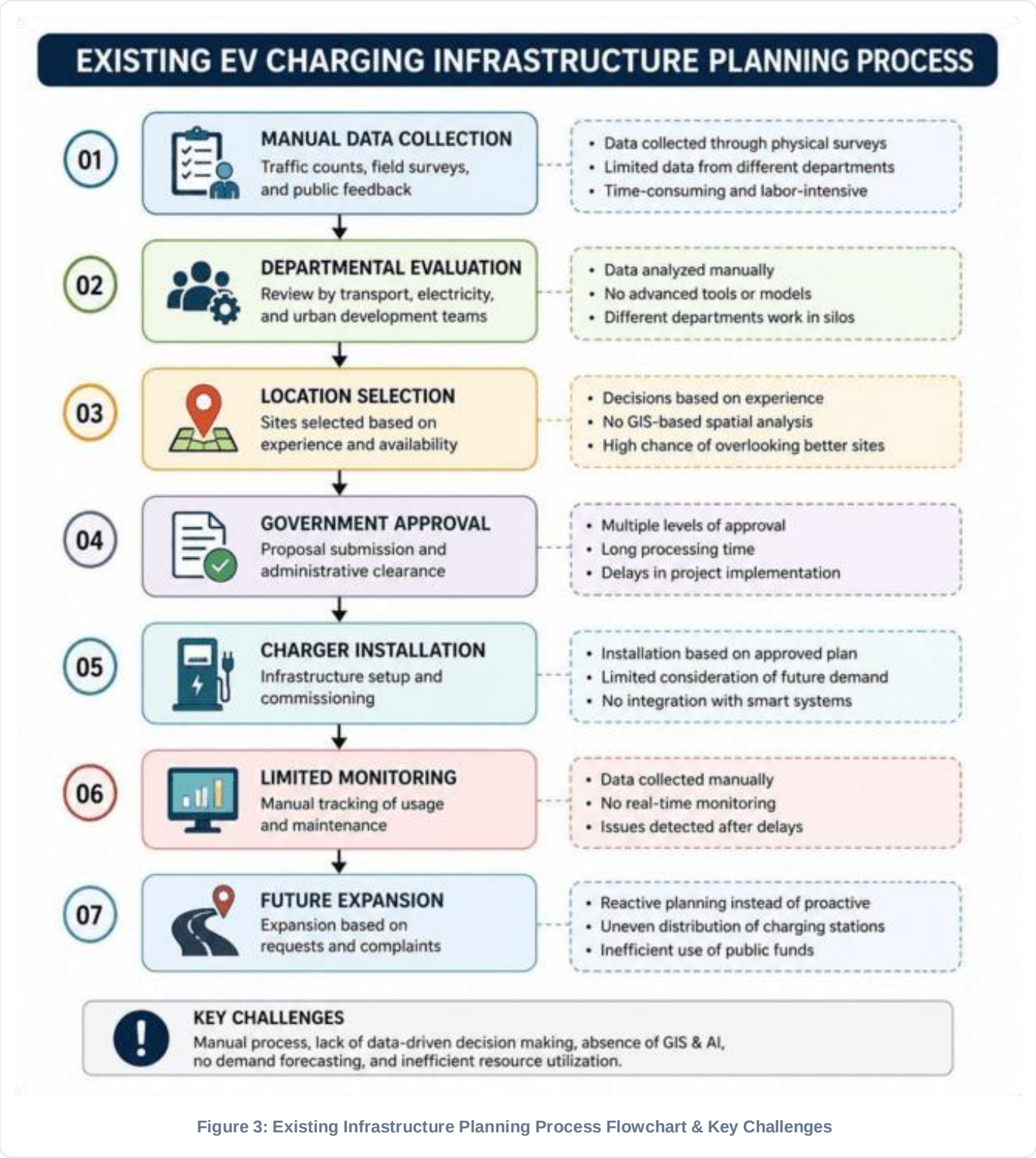


Figure 3: Existing Infrastructure Planning Process Flowchart & Key Challenges

8. PROPOSED SOLUTION & UNIQUENESS

EVGrid AI is an AI- and GIS-based decision support system developed to help Kopargaon Taluka plan EV charging infrastructure scientifically. By merging traffic volume, population density, tourism index, grid capacity, and commercial activity, the platform calculates weighted suitability scores.

Existing Planning	EVGrid AI System
Manual surveys and field inspections	AI-based location recommendations
No GIS support	GIS heat maps & spatial analysis
No future demand prediction	AI-powered demand forecasting (2027–2035)
Static planning approach	Dynamic & data-driven planning
One-time decision making	Continuous optimization with updated data
Limited grid analysis	Grid feasibility & capacity assessment
Manual budget allocation	Automated budget optimizer

9. INNOVATION HIGHLIGHTS

- **AI Suitability Score:** Ranks locations based on traffic, population, tourism, and grid capacity.
- **GIS Heat Maps:** Identifies high-demand zones through interactive spatial mapping.
- **Budget Optimizer:** Recommends optimal charger types and counts within specified budget caps.
- **Carbon Emission Calculator:** Measures environmental impact (CO2 reduced, fuel saved).
- **Demand Forecasting Engine:** Projects charging needs for 2027, 2030, and 2035.
- **Government Decision Dashboard:** Centralized interface with instant PDF report generation.

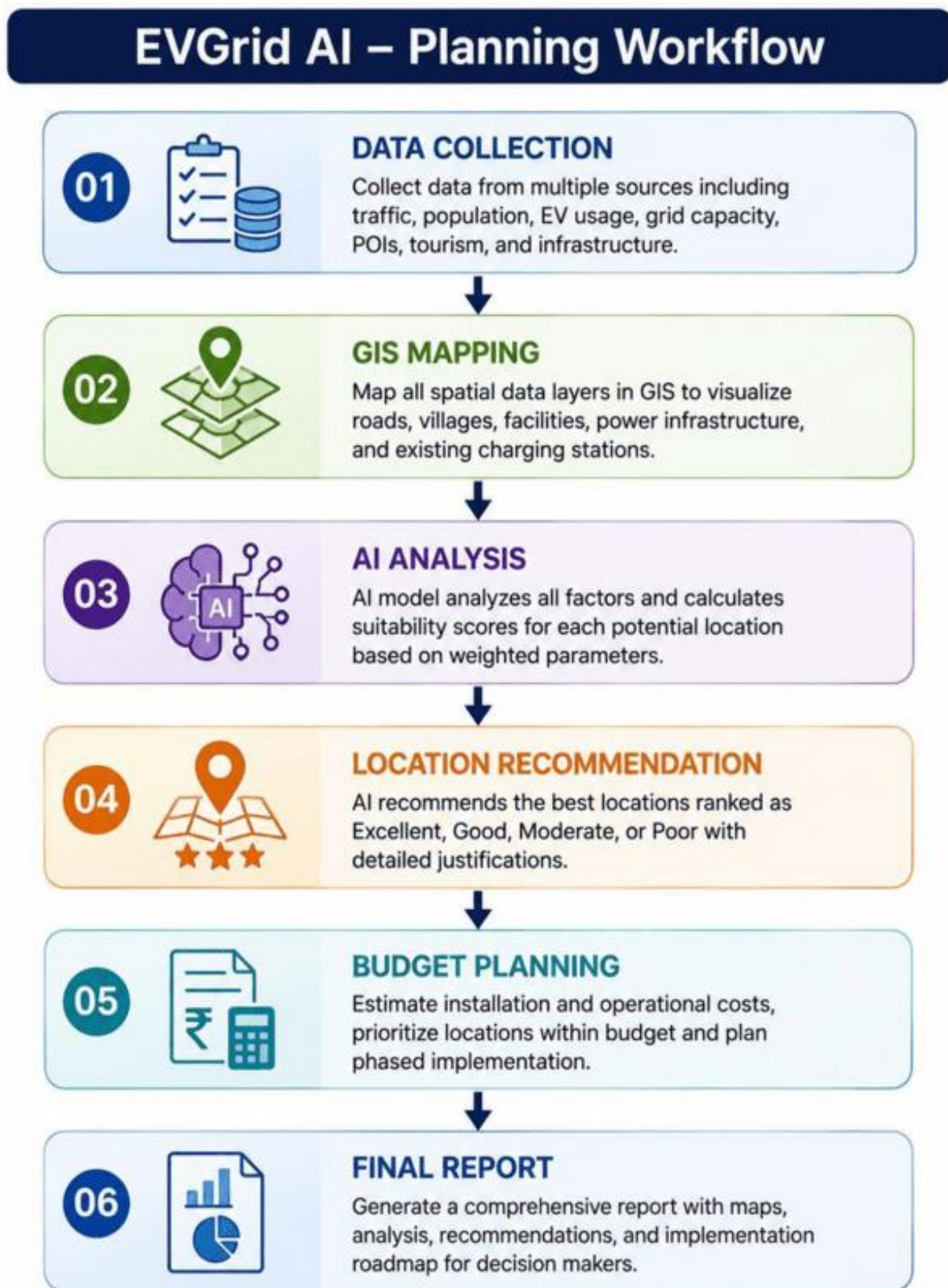


Figure 4: EVGrid AI Planning Workflow (Data Collection to Final Decision Report)

11. FUNCTIONAL REQUIREMENTS & UI LAYOUT

The EVGrid AI platform consists of 10 integrated modules designed for seamless government operations.



Figure 5: Detailed Functional Modules & System User Interface Features

12. NON-FUNCTIONAL REQUIREMENTS

- **Performance:** Dashboard loading time < 3 seconds; Map rendering < 5 seconds.
- **Security:** Role-based access control (RBAC), password encryption, and secure APIs.
- **Reliability:** 99% operational uptime and reliable spatial query execution.
- **Scalability:** Flexible architecture scalable from Taluka to District and State levels.
- **Usability:** Intuitive, clean UI tailored for government decision-makers.

13. SYSTEM ARCHITECTURE

The architecture connects government officers to an interactive web dashboard powered by GIS, AI, and analytics engines built over a PostgreSQL/PostGIS spatial database

SYSTEM ARCHITECTURE

EVGrid AI – Smart EV Infrastructure Planning

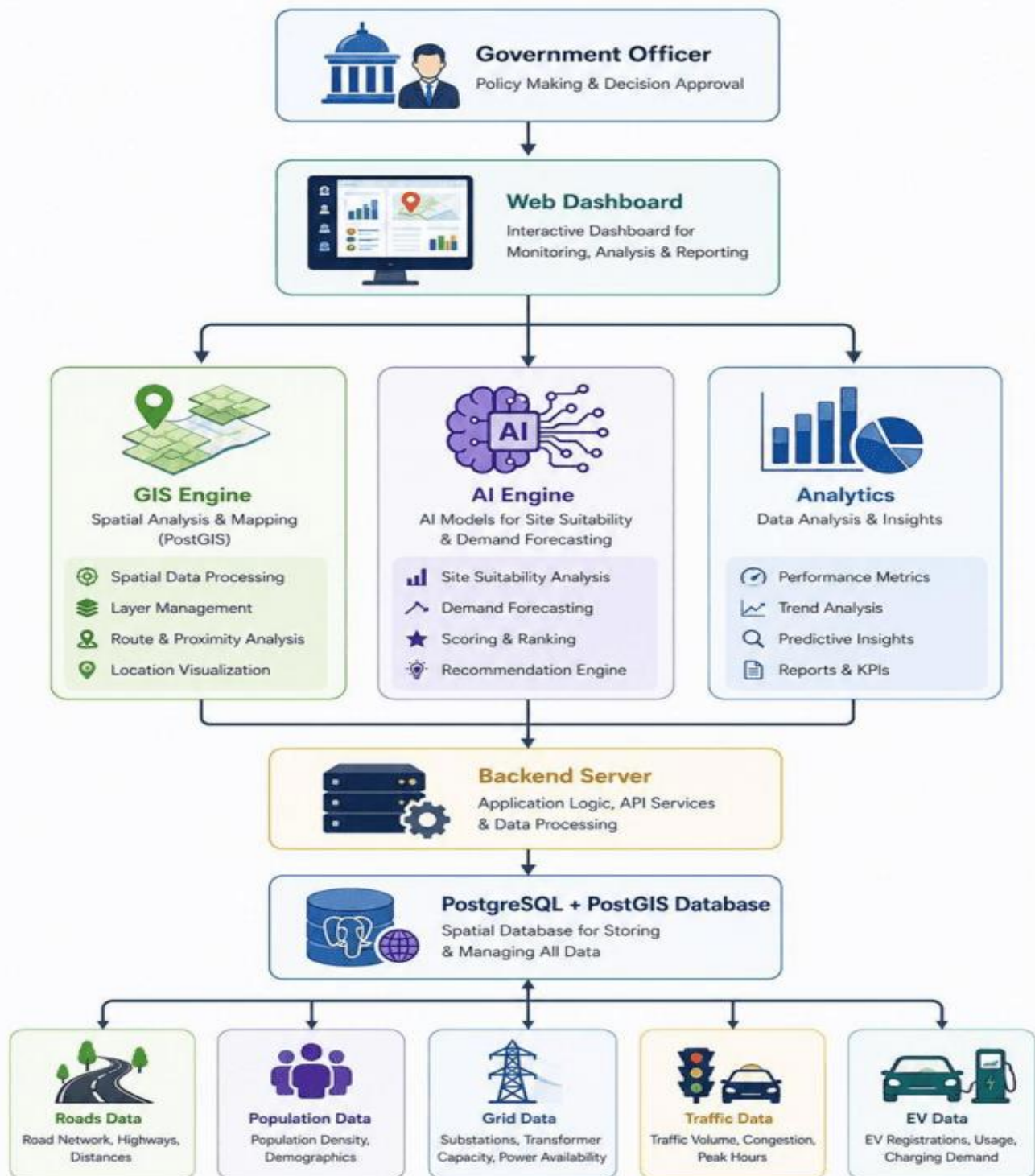


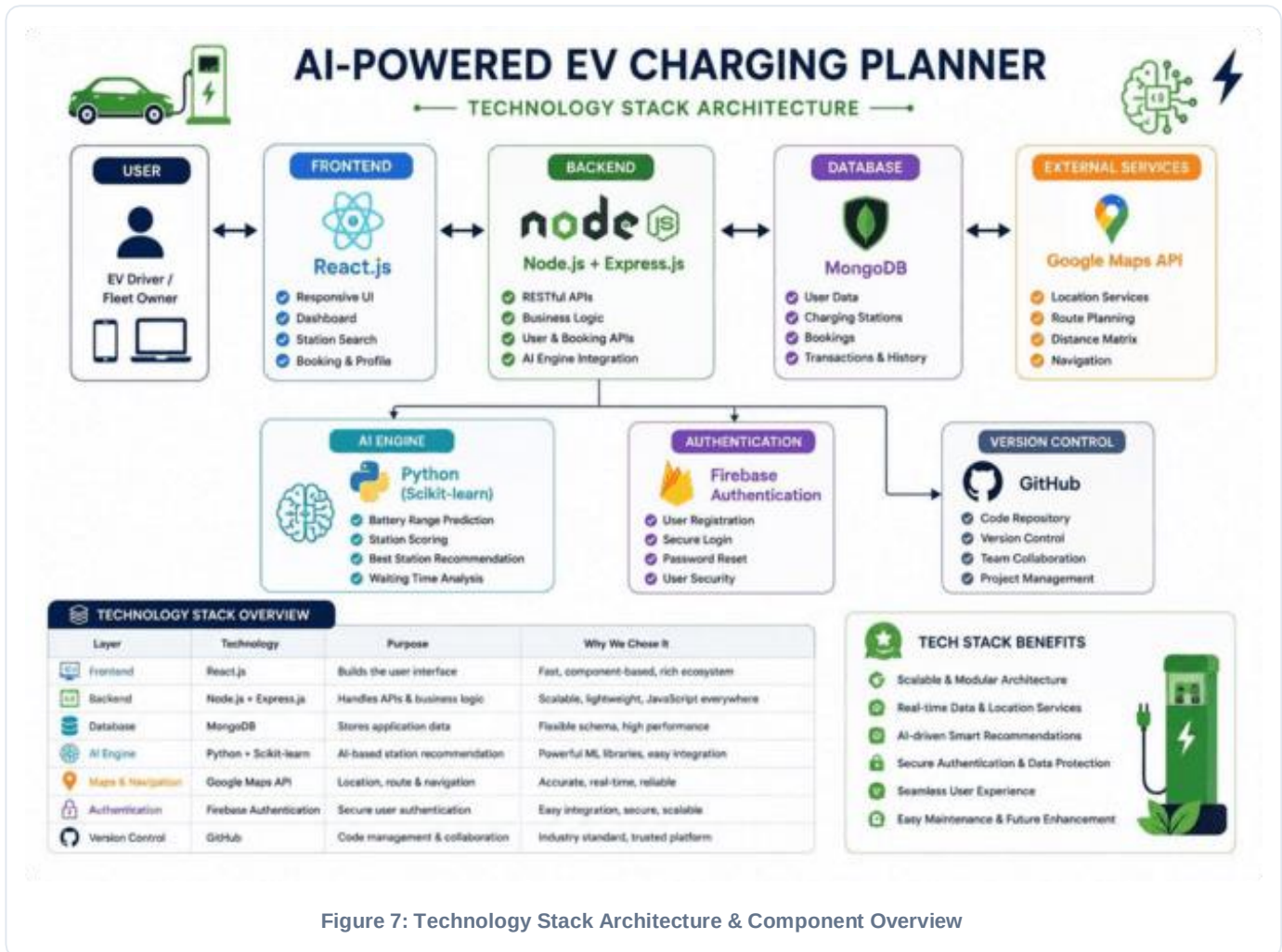
Figure 6: EVGrid AI System Architecture Diagram

14. DATABASE DESIGN

The system stores and manages data through structured relational and spatial tables:

- **Table 1: Locations** (Location ID, Village Name, Latitude, Longitude, Population, Traffic Density, Parking Capacity, Tourist Score, Commercial Score)
- **Table 2: Charging Stations** (Station ID, Name, Latitude, Longitude, Charger Type, Capacity, Status)
- **Table 3: Electricity Grid** (Substation Name, Transformer Capacity, Available Power, Grid Load)
- **Table 4: Demand Forecast** (Year, Predicted EVs, Charging Demand, Peak Hour, Suggested Chargers)

15. TECHNOLOGY STACK ARCHITECTURE



16. AI EVALUATION PARAMETERS & SCORE CATEGORIES

Evaluation Parameter	Weight Percentage (%)
Traffic Density	25%
Population Density	20%
Grid Capacity & Proximity	20%

Evaluation Parameter	Weight Percentage (%)
Parking Availability	10%
Commercial Area Activity	10%
Tourist / Pilgrim Flow	10%
Future EV Growth Potential	5%

Suitability Score Categories:

- **80 – 100:** Excellent Location (Immediate Priority)
- **60 – 79:** Good Location (Phase 1/2 Deployment)
- **40 – 59:** Moderate Location (Secondary Phase)
- **Below 40:** Poor Location (Not Recommended)

17. PROPOSED LOCATIONS & FIELD PHOTOGRAPHS

Field survey site assessments conducted across Kopargaon Taluka identified five primary priority locations



Proposed Priority Hubs Overview:

- 1. **Kopargaon Railway Station:** Fast DC Chargers (Serves tourist vehicles, taxis, available parking).
- 2. **MSRTC Bus Stand:** Fast + Slow Chargers (Serves public buses, commuters, commercial vehicles).
- 3. **Main Commercial Market:** Slow AC Chargers (Serves shoppers, commercial vehicles, long-stay parking).
- 4. **Shirdi Highway Junction:** High-speed Fast Chargers (Serves highway traffic & pilgrims).
- 5. **Hospital Area:** Emergency charging (Serves EV ambulances and emergency public access).

18. EXPECTED OUTPUTS & BUDGET BREAKDOWN

Project Phase / Component	Estimated Budget Range
Software Prototype & Modeling	₹3,000 – ₹10,000
Pilot Deployment (Selected Locations)	₹25 Lakh – ₹40 Lakh
Taluka-wide Platform Implementation	₹90 Lakh – ₹1.2 Crore
Full Infrastructure Setup (24+ Stations)	₹5 Crore – ₹6 Crore

19. USER FLOW DIAGRAM (APP FLOW)



AI-Powered EV Charging Planner

USER FLOW DIAGRAM

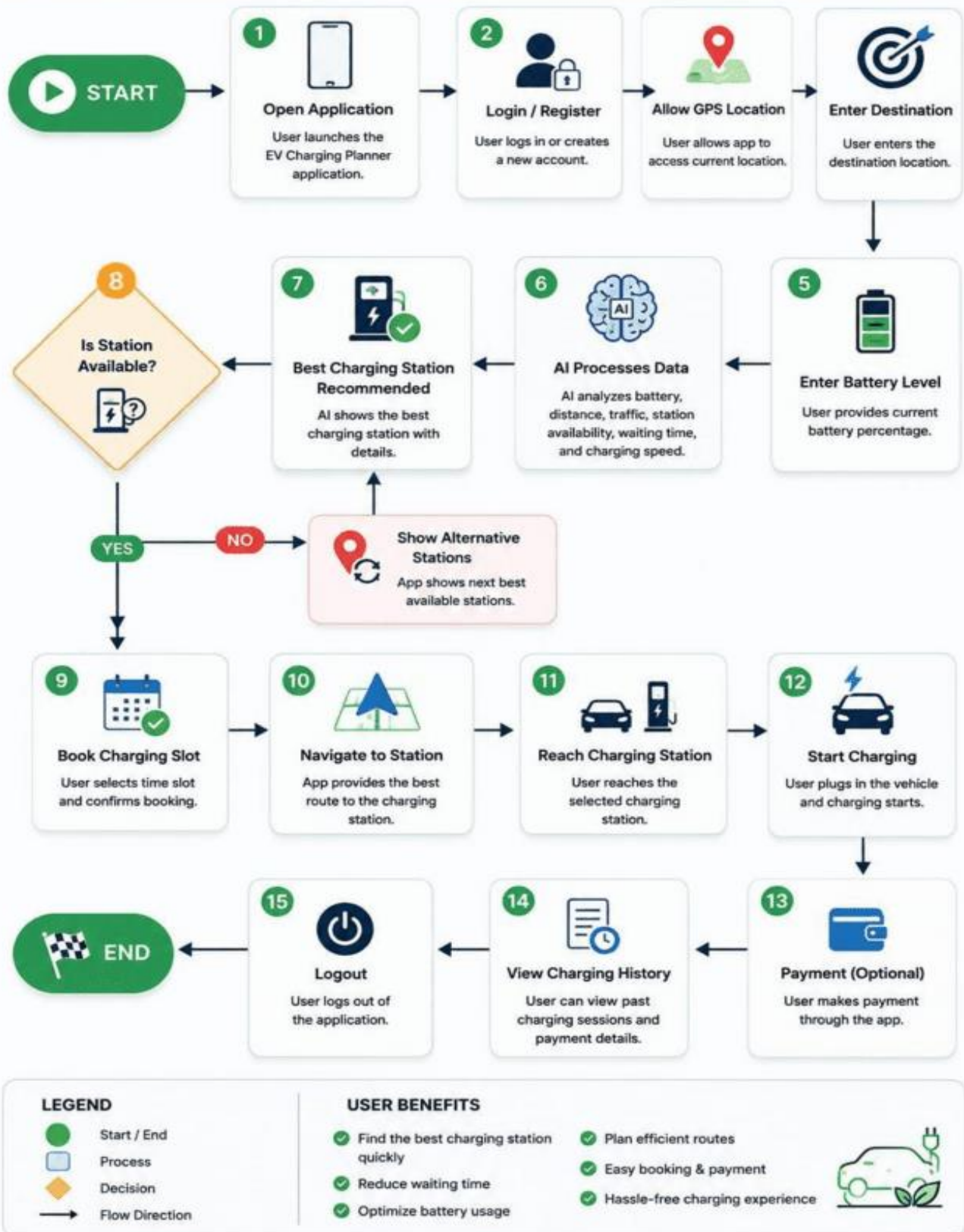


Figure 8: End-User & Driver Navigation Application Flow Diagram

20. IMPLEMENTATION ROADMAP



Figure 9: Implementation Roadmap - From Data to Scalable Impact

21. RISK ANALYSIS & SDG ALIGNMENT

Risk Mitigation Strategies

Identified Risk	Mitigation Strategy
Incomplete or outdated data	Continuous data pipelines, API syncing, and periodic surveys
Grid overload during peak hours	AI smart load balancing and solar-battery storage integration
Budget constraints	Phase-wise deployment prioritizing high-suitability score hubs

Alignment with UN Sustainable Development Goals (SDGs)

- **SDG 7:** Affordable and Clean Energy
- **SDG 9:** Industry, Innovation, and Infrastructure
- **SDG 11:** Sustainable Cities and Communities
- **SDG 13:** Climate Action

22. FUTURE SCOPE & CONCLUSION

Future Scope

Future extensions include real-time IoT station status monitoring, MSEDCL grid dynamic balancing, dynamic slot reservation apps, vehicle-to-grid (V2G) power sharing, and predictive maintenance algorithms for EV charging hardware.

Conclusion

EVGrid AI is an AI-driven decision support platform designed to help government authorities scientifically plan EV charging infrastructure. By combining Artificial Intelligence, GIS mapping, demand forecasting, electricity grid analysis, and budget optimization, the platform enables data-driven planning instead of manual decision-making. Developed by Team Nirdhar, this solution supports Kopergaon's transition toward sustainable mobility while providing a scalable model that can be adapted across Maharashtra and India.

23. REFERENCES

1. Government of India - National Electric Mobility Mission Plan (NEMMP)
2. Maharashtra Electric Vehicle Policy 2021
3. Ministry of Heavy Industries (Government of India) – EV initiatives & FAME India Scheme
4. MSEDCL (Maharashtra State Electricity Distribution Company Ltd.) - Grid & Power Data
5. NITI Aayog - Sustainable Transportation & Mobility Reports
6. OpenStreetMap (OSM) - Geospatial Mapping Data
7. Ministry of Road Transport and Highways (MoRTH) – Road & Traffic Statistics
8. Government of Maharashtra - Smart City Initiatives